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Reliability and Maintenance Considerations in Process Design under Uncertainty

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Abstract.

In this paper, we describe recent theoretical and algorithmic developments aiming at the integration of maintenance optimization in optimal life-cycle process design and development under uncertainty. Based on a mathematical description for the process model, information regarding the reliability characteristics of equipment components and type and cost of different maintenance policies for each component, we propose an optimization formulation featuring an expected profit objective function, taking into account the transitions between the different states of the process system (due to equipment failure and corrective maintenance). Different types of preventive maintenance strategies are discussed together with their resulting mathematical problems, which correspond to mixed-integer nonlinear optimization formulations involving implicitly defined integral terms. By exploiting properties of the maintenance problem, a series of nonlinear optimization problems of reduced size is solved instead, thereby avoiding the direct solution of the original formulations. The implications of maintenance optimization on process design are also studied. This is achieved by comparing two approaches, a sequential vs. a simultaneous strategy, for simultaneous design and maintenance optimization. It is shown that the incorporation of maintainability criteria provides new directions in proposing and identifying optimal, over a life cycle, process designs. © 1998 Elsevier Science Ltd. All rights reserved.

Keywords: process design, reliability, maintenance, process uncertainty, optimization.

INTRODUCTION

Although the need for integrating different operability criteria in the study of chemical process and operation has been widely recognised (Van Rijn, 1987; Grievink *et al.*, 1993), at the design stage of process plants, operability considerations accounting for the plant's life cycle, such as flexibility, reliability and maintainability are not typically included. The main reason for this is that currently there is a lack of an integrated design framework enabling process engineers to look at the various operability factors in conjuction with cost in a systematic and quantified way during process design development. On the other hand, traditional availability assessment of process systems, provides quantitative information on the characteristics of the maintenance policy to be followed, based on the system reliability model but without taking into account detailed process interactions, the process model complexity, and the uncertainty which may be involved in a number of model and proccess parameters. Gradually, general mathematical frameworks have started to appear (Straub and Grossmann, 1993; Thomaidis and Pistikopoulos, 1995). Most studies, however, have not explicitly addressed maintenance considerations in the presence of uncertainty. On the other hand, typical studies on maintenance optimization (Tseng, 1996; Alkhamis and Yellen, 1995) do not take into account the important interactions of maintenance and the process system design and operation in the presence of uncertainty.

In this paper we present a novel maintenance optimization algorithm in the presence of uncertainty. Methods for integrating process design and maintenance optimization by exploring their interactions during the life-cycle of a process are also presented.

MATHEMATICAL FOUNDATIONS AND MODELS

Consider a process system, such as the one shown in Figure 1, comprising four components. Over a time horizon H, given that equipment is subject to failure and corrective maintenance, the system may reside in any of the sixteen states with a corresponding probability $P_k(t)$. Associated with each system state k, there is an operational envelope (feasible operating region, FOR_k) defined by the appropriate process model (equality and inequality constraints, see Pistikopoulos, 1995). Uncertainty is also involved in

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the process model; for example, the supply of raw material A and the demand of product C are described by probability distribution functions (pdf).

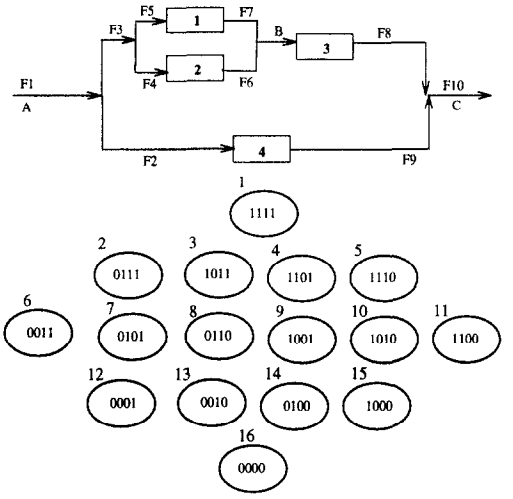


Figure 1: System with four components and sixteen states

Given a revenue rate function, an expected revenue function can be defined as (for mathematical notation, see Nomenclature section):

$$ER_H = \int_H \sum_{k \in S} P_k(t) \cdot ER_k dt \qquad (1)$$

where

- $ER_k = \int_{\theta \in FOR_k} r(d,\theta) j(\theta) d\theta$ is the expected profit rate function of state k
- $r(d,\theta)$ is the revenue rate function of the system as a function of design variables d and uncertain parameters θ.
- $j(\theta)$ is the pdf describing the vector of uncertainty

Note that in (1) process design, uncertainty and maintenance characteristics are all included. The term ER_k depends on the system configuration, the values of the design variables and the realization of uncertain parameters. On the other hand, the probability of being in each system state k at any time t, $P_k(t)$, is determined by the probability of each component j of the system being up (or down) at t, which in turn depends on the implementable maintenance policy. A model for describing the maintenance policy follows.

Maintenance policy model

Let $R_{1,j}(t)$ be the initial (before any maintenance is performed) reliability function of component j as a function of time, and t_i, $\forall i \in [2..N-1]$ the time instants at which we perform maintenance to any of the components according to a maintenance schedule.

First, we make the following assumptions:

- at each maintenance stage, maintenance is performed on only one of the units in the following way:
  - *corrective*, if the unit is down
  - *preventive*, if it is operable.

  This assumption is common in the literature (Tseng, 1996) and valid in many real cases
- all maintenance is of an AGAN type; similar expressions can be derived for any other maintenance type
- all failure events are independent.

Suppose, now, that our system is in the time interval between t_{i-1} and t_i and let $R_{i-1,j}(t)$ denote the reliability function of unit j during this interval. We introduce two types of optimization variables: t_i, which are *continuous* variables, representing the time instant of maintenance action $i-1$, and $u_{i,j}$, which are *0-1* variables defined as :

$$u_{i,j} = \begin{cases} 1 & \text{if maintenance action is performed on component j at time instant } t_i \\ 0 & \text{else} \end{cases}$$

The reliability of component j after t_i will then be

$$R_{i,j}(t) = (1-u_{i,j}) \cdot R_{i-1,j}(t) + u_{i,j} \cdot R_{1,j}(t-t_i') \quad (2)$$

where $t_i' = t_i + \tau_i$, τ_i being the duration of maintenance action i-1.

Eq. (2) describes the following two cases:

- $u_{i,j} = 0$: then maintenance at t_i has been planned to take place on a component other than j. Hence, component j retains its reliability characteristics during the previous maintenance interval

$$R_{i,j}(t) = R_{i-1,j}(t)$$

- $u_{i,j} = 1$: then maintenance action (corrective or preventive) at t_i is performed on unit j which, according to our assumptions, becomes AGAN

$$R_{i,j}(t) = R_{1,j}(t-t_i')$$

Function $R_{1,j}(t-t_i')$ is simply function $R_{1,j}(t)$ shifted in time by t_i'.

With the use of Equation (2) we can establish analytical expressions for the reliability functions of all units *during* and *after* each maintenance action. For a detailed analysis on how to derive such expressions see Vassiliadis and Pistikopoulos(1998).

Overall optimization model

Based on the modelling and mathematical foundations described above and equations (1) and (2), the following general mathematical model (P1) is proposed for the simultaneous optimization of design and maintenance in the presence of uncertainty.

$$\textbf{(P1)} \quad \max_{N,u_{i,j},t_i,d,z_k,x_k} \sum_{i=1}^{N-1} \int_{t_i+\sum_{j=1}^{M} u_{i,j}\tau_j}^{t_{i+1}} \cdot \sum_{k\in S} P_{i,k}(t)\cdot ER_k$$

$$+\sum_{i=2}^{N-1} \int_{t_i}^{t_i+\sum_{j=1}^{M} u_{i,j}\tau_j} \sum_{k\in S} \bar{P}_{i,k}(t)\cdot ER_k$$

$$-m.cost(u_{i,j},t_i,N)$$

$$-d.cost(d_j)$$

s.t.

$$h_k(d,z_k,x_k,\theta)\le 0,\quad d\in D, z_k\in Z, x_k\in X, \forall k\in S$$

$$g_k(d,z_k,x_k,\theta)\le 0, \qquad (3)$$

$$m.cost(u_{i,j},t_i,N)=\sum_{\lambda=2}^{N-1}\sum_{j=1}^{M} u_{\lambda,j}\cdot$$

$$\cdot\Big[P(F_{\lambda,j})C_{corr,j}+(1-P(F_{\lambda,j}))C_{prev,j}\Big] \qquad (4)$$

$$d.cost(d_j)\le LIMIT,\quad d_j\in D,\forall j \qquad (5)$$

$$\sum_{j=1}^{M} u_{\lambda,j}=1,\forall\lambda\in[2..N-1]$$

$$t_i+\sum_{j=1}^{M} u_{i,j}\cdot\tau_j\le t_{i+1},\quad \forall i\in[1..N-1] \qquad (6)$$

$$u_{1,j}=1,\forall j\in[1..M]$$

N is an integer variable denoting that the total number of maintenance actions is N-2. The expressions in (3) represent the process constraints h_k and g_k as a function of design (d), control (z_k), state variables (x_k) and uncertain parameters (θ). (4) represents total maintenance costs, while the term $P(F_{\lambda,j}) = 1 - R_{\lambda-1,j}(t_\lambda)$ denotes the probability of unit j being down when maintenance action λ occurs. (5) imposes an upper limit for the total design cost. The inequality, (6), ensures that the time instant of the λth maintenance action follows that of $(\lambda-1)^{th}$.

Problem (P1) corresponds to a mixed-integer nonlinear optimization formulation involving discrete decisions (number and sequence of maintenance actions) and integral terms with integrands defined implicitly (as a function of 0-1 variables). The direct solution of (P1) is therefore very complex - an effective two step solution strategy is proposed in the following section which avoids the direct solution of (P1) by exploiting its mathematical structure and establishing reliability properties.

MAINTENANCE OPTIMIZATION ALGORITHM

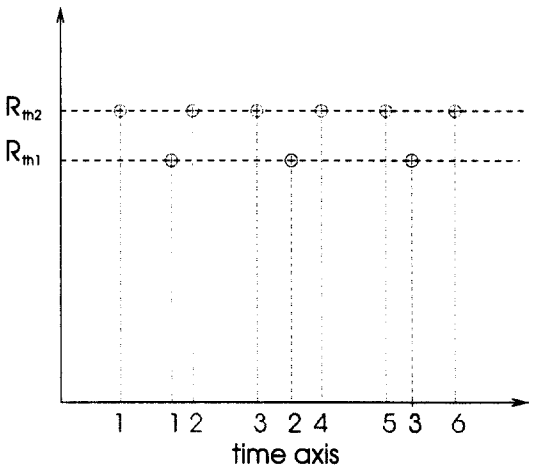


Figure 2: Reliability threshold values

If the design variables (and process structure) are fixed, problem (P1) essentially reduces to a maintenance optimization problem with the aim to determine (i) *the number and sequence of maintenance actions*, i.e. the values of N and $u_{i,j}$, and (ii) *the maintenance time instants*, i.e. the values of the continuous variables t_i. We propose the following two-step strategy:

Step 1: Determining an initial sequence

Each maintenance time instant corresponds to a reliability value of the unit to be maintained (due to the monotonicity of the reliability function). Therefore, an alternative way to identify the maintenance time instants is to determine an optimal set of reliability *threshold* values. We seek to depict such threshold values by considering them fixed for each unit throughout the time horizon. A maintenance

action is executed on an equipment unit each time its reliability function reaches this threshold value. This is schematically shown in Fig 2, for a two component system, where nine maintenance actions take place (six for component 2 and three for component 1) with a maintenance sequence of 212212212.

These constant (over time) reliability threshold values for each unit j, $\mathcal{R}_{th,j}$, are determined from the solution of an optimization problem derived from (P1) by expressing all terms as function of the threshold values. These are summarised below (see Vassiliadis and Pistikopoulos, 1998)

- for fixed design d, the term ER_k can be separately obtained from the solution of the following optimization problem for each state k (see Thomaidis and Pistikopoulos, 1995):

$$ER_k = \max_{z_k, x_k} \int_{\theta \in FOR_k} r_k(\theta) j(\theta) d(\theta)$$

$$s.t. \quad h_k(d, z_k, x_k, \theta) \leq 0, \quad \forall k \in S$$
$$g_k(d, z_k, x_k, \theta) \leq 0, \quad \forall k \in S \qquad (7)$$

- $H_k(R_{th,j})$: this term represents the expected time that the system may spend in state k as function of the reliability thresholds. For example for the fully operable state of the system, assuming exponentially distributed unit reliability functions, we have

$$H_1(R_{th,j}) = H \cdot (1 + \sum_j MTTF_j \cdot ln(R_{th,j})) \quad (8)$$

Similar expressions are derived for all system states.

- $E\{Pr_k(R_{th,j})\}$: it represents the expected value of the probability of the system being in state k throughout the time horizon. Since the time of failure of each unit is a random variable, the reliability functions as well as the state probabilities are also random variables. For an exponentially distributed reliability function the probability distribution function of R is equal to

$$f(R|R_{th,j} < R \leq 1) = \frac{1}{1 - R_{th,j}}$$

and the expected reliability value is then given by

$$E(R) = \frac{1 + R_{th,j}}{2}$$

The expected value of the probability of occurence of each state can also be evaluated. For the fully operable state, this value is given by

$$E\{Pr_k(R_{th,j})\} = \frac{1 + R_{th,1}}{2} \cdot \frac{1 + R_{th,2}}{2} \cdot \ldots \cdot \frac{1 + R_{th,n}}{2} \qquad (9)$$

since failures are assumed independent. Similar expressions can be derived for all system states.

- $m.cost(R_{th,j})$: this terms represents the total maintenance cost as a function of the reliability threshold values. From Eq. (4) , assuming exponentially distributed unit reliability functions, the following expression can be obtained:

$$m.cost = -\sum_j \frac{H}{MTTF_j \cdot ln(R_{th,j})} \cdot \left[R_{th,j} C_{prev,j} + (1 - R_{th,j}) C_{corr,j}\right] \qquad (10)$$

Substituting the above expressions, (7)-(10), in (P1) results in the following nonlinear optimization problem (P2):

$$\textbf{(P2)} \quad \max_{\mathcal{R}_{th,j}} \sum_{k \in S} H_k(R_{th,j}) \cdot E\{Pr_k(R_{th,j})\} \cdot ER_k - m.cost(R_{th,j})$$

$$s.t. \quad 0 \leq \mathcal{R}_{th,j} \leq 1, \quad \forall j \in [1..M]$$

The solution of (P2) then provides the optimal threshold values $R_{th,j}$, from which the number of maintenance actions $\bar{N}$ and an initial sequence of maintenance actions ,$\bar{u}_{i,j}$ can be determined.

Step 2: Finding an optimal solution

(a) With fixed $\bar{N}$ and $\bar{u}_{i,j}$, problem (P1) reduces to a standard nonlinear optimization problem (P3):

$$\textbf{(P3)} \quad \max_{t_i} \sum_{i=1}^{\bar{N}-1} \int_{t_i + \sum_{j=1}^{M} \bar{u}_{i,j}\tau_j}^{t_{i+1}} \sum_{k \in S} P_{i,k}(t) ER_k + \sum_{i=2}^{\bar{N}-1} \int_{t_i}^{t_i + \sum_{j=1}^{M} \bar{u}_{i,j}\tau_j} \sum_{k \in S} \bar{P}_{i,k}(t) ER_k - \sum_{\lambda=2}^{\bar{N}-1} \sum_{j=1}^{M} \bar{u}_{\lambda,j} \left[P(F_{\lambda,j}) C_{corr,j} + (1 - P(F_{\lambda,j})) C_{prev,j}\right]$$

$$s.t. \quad t_i + \sum_{j=1}^{M} \bar{u}_{i,j} \cdot \tau_j \leq t_{i+1}, \quad \forall i \in [2..\bar{N} - 1]$$

The solution of (P3) yields the maintenance time instants t_i, $i \in [2..\bar{N} - 1]$.

(b) If some of the inequality constraints in (P3) are active, i.e.

$$t_i + \sum_{j=1}^{M} \bar{u}_{i,j} \cdot \tau_j = t_{i+1}, \quad \forall i \in [1..\bar{N}-1]$$

we go back to step 2(a) and check whether changing the sequence $\bar{u}_{i,j}$ of the maintenance actions involved in this constraint will improve the solution of (P3).

(c) If there are no active constraints or no better solution is obtained, stop.

While this two-step strategy does not theoretically guarantee to identify the optimal solution of problem (P1) in a strict mathematical sense but only a valid lower bound solution, this approximation is often very close to the exact solution (as further discussed in Vassiliadis and Pistikopoulos, 1998).

SIMULTANEOUS DESIGN AND MAINTENANCE OPTIMIZATION

Two optimization strategies, a sequential and a simultaneous, are proposed and compared in this section for studying the interactions of design decisions and maintenance aspects. As shown in Fig. 3, in the sequential approach, a design optimization step is first performed followed by the implementation of the maintenance optimization algorithm (described in the previous section) for the optimal design depicted. On the other hand, in the simultaneous approach the design optimization step is incorporated within problem (P2) where the optimal maintenance time instants are depicted in a second step. A brief description of the two strategies follows, whereas, for simplicity in the presentation, aspects of uncertainty are not included.

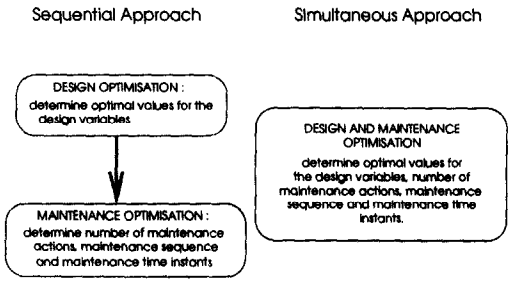


Figure 3: Sequential vs. simultaneous approach

Sequential optimization

The sequential approach comprises three steps: in the first step an optimal design d is obtained without accounting for life-cycle maintenance aspects, in the next two steps maintenance optimization is performed (for the optimal design d) over a time horizon H. A summary of the steps follows:

Step1: For the fully operable state, solve the following optimization problem:

$$\textbf{(P4)} \qquad \max_{d,z,x} r(d,z,x)$$

$$s.t. \qquad h(d,z,x) = 0$$
$$g(d,z,x) \leq 0$$

to obtain the optimal vector of design variables d.

Step2: For the optimal d, solve problem (P2) to obtain the optimal reliability threshold values, $R_{th,j}$; the number of maintenance actions N and an initial maintenance sequence $\bar{u}_{i,j}$ are also determined.

Step3: Solve problem (P3) to obtain an optimal maintenance sequence and the optimal maintenance time instants $t_{i,j}$ (see step 2, maintenance optimization section).

Simultaneous optimization

The simultaneous approach involves two steps. In the first step, an optimal design d is simultaneously determined with the optimal equipment reliability threshold value, while in the second step an optimal maintenance sequence is obtained.

Step1: By incorporating design considerations, problem (P2) is recasted as follows:

$$\max_{d_j, z_k, x_k, R_{th,j}} \sum_{k \in S} H_k(R_{th,j}) \cdot Pr_k(R_{th,j}) \cdot ER_k(d, z_k, x_k)$$

$$\textbf{(P5)} \qquad - m.cost(R_{th,j}) - d.cost(d_j)$$

$$s.t. \qquad h(d, z_k, x_k) = 0$$
$$g(d, z_k, x_k) \leq 0$$
$$0 \leq R_{th,j} \leq 1 \quad , \forall j$$
$$d.cost(d_j) \leq LIMIT$$

The solution of (P5) yields (i) the optimal vector of design variables d, and (ii) the optimal reliability threshold values $R_{th,j}$ from which the number of maintenance actions N, and an initial maintenance sequence $\bar{u}_{i,j}$ can be defined.

Step2: Solve problem (P3) to obtain an optimal maintenance sequence and the corresponding optimal maintenance time instants $t_{i,j}$.

EXAMPLES

Example 1: Maintenance optimization under uncertainty (fixed design).

The chemical complex shown in Fig. 1 (Straub and Grossmann,1990) converts species A to C either

	Mass Balance	Specifications
F = flowrates	$F_1 = F_2 + F_3$	$F_5 \leq d_1 y_1$
d = capacity	$F_3 = F_4 + F_5$	$F_4 \leq d_2 y_2$
S = supply	$F_{10} = F_8 + F_9$	$(F_6 + F_7) \leq d_3 y_3$
D = demand	$F_7 = \alpha_1 F_5$	$F_2 \leq d_4 y_4$
	$F_6 = \alpha_2 F_4$	$F_1 \leq S$
	$F_8 = \alpha_3 (F_6 + F_7)$	$F_{10} \geq D$
	$F_7 = \alpha_4 F_2$	

Table 1: Chemical complex data and equations-Example 1

through the production of the intermediate product B or by direct conversion of A to C. The supply of raw material A and the demand of product C are considered as continuous uncertain parameters, described by normal pdf's. The variables and parameters, along with the equations used to model this system are given in Table 1. The operating, reliability and maintenance cost characteristics of the process are provided in Table 2.

The NLP formulations were solved using the GAMS modelling package (Brooke *et al.*, 1988). A time horizon of $H = 500$ time units has been considered.

Unit	MTTF	$C_{prev,j}$	$C_{corr,j}$	α	d
1	1000	2000	10000	0.92	5
2	1000	2000	10000	0.9	5
3	1000	2000	10000	0.85	7
4	1000	2000	10000	0.75	9

Table 2: Reliability and operating characteristics-Example 1

We have evaluated the revenue rate for each one of the nine operable states of the system by integrating the pdf of the profit rate function over the FOR of each state using a Gaussian integration scheme with ten quadrature points for both continuous uncertainties similar to the one presented in Pistikopoulos and Ierapetritou, (1995). The results are summarized in Table 3 where y_i is a binary variable which takes the value of 1 if component i is operable.

By applying the steps of the maintenance optimization algorithm, the following results are obtained.

Step 1: The optimal reliability thresholds for each component are first determined (from the solution of (P3), with which the corresponding initial maintenance sequence can be derived, as shown in Table 4.

Note that the optimal reliability threshold values are different even though reliability and maintenance characteristics have been assumed identical for all units. This is due to the different relative importance of each component to the expected revenue of the process. We also note that the thresholds values of units 3 and 4 are significantly bigger than those of units 1 and 2, since they "participate" in more and more profitable states of the system.

$S^N = 12$	$\sigma_S = 1$	Cost rate of A = 40
$D^N = 7$	$\sigma_D = 1$	Revenue rate from C = 200

State	y_1	y_2	y_3	y_4	ER_k
1	1	1	1	1	1147
2	0	1	1	1	1125
3	1	0	1	1	1133
4	1	1	0	1	343
5	1	1	1	0	114
6	0	0	1	1	343
7	0	1	0	1	343
8	1	0	0	1	343
9	0	0	0	1	343

Table 3: Uncertainty and revenue characteristics-Example 1

Unit	$\mathcal{R}_{th}$	M.Instants	M.Actions
1	0.730	315	1
2	0.710	343	1
3	0.903	102, 204, 306, 408	4
4	0.919	85, 170, 255, 340, 425	5
Init.Maint.Sequence		4,3,4,3,4,3,1,4,2,3,4	

Table 4: Results of step 1-Example 1

It.	M.Sequence
1	4,3,4,3,4,*3,1*,4,2,3,4
2	4,3,4,3,*4,1*,3,4,2,3,4
3	4,3,4,3,1,4,3,*4,2*,3,4
4	4,3,4,3,1,4,3,2,4,3,4

M.Instants	Profit
82,83,164,166,245,252,253,322,323,369,406	4.83E5
73,101,145,205,218,219,303,304,305,39,397	4.84E5
79,97,156,197,198,233,298,311,312,394,400	4.843E5
84,94,167,191,192,245,288,291,327,389,408	4.845E5

Table 5: Results of Step 2-Example 2

Step 2: With fixed maintenance sequence (4,3,4,3,4,3,1,4,2,3,4), as determined in Step 1, the solution of (P3) provides the optimal maintenance time instants, as shown in Table 5.

At the solution, the inequality constraint regarding certain maintenance actions becomes active. By reversing the order and resolving problem (P3) an optimal solution is obtained in four iterations.

Example 2: Design and maintenance optimization

Both sequential and simultaneous optimization

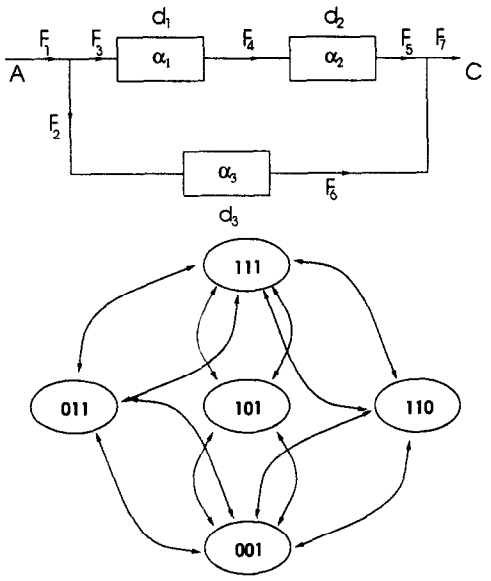


Figure 4: Multistate System with three components

	Mass Balance	Specifications
F = flowrates	$F_1 = F_2 + F_3$	$F_3 \leq d_1 y_1$
d = capacity	$F_4 = \alpha_1 F_3$	$F_4 \leq d_2 y_2$
S = supply	$F_5 = \alpha_2 F_4$	$F_2 \leq d_3 y_3$
D = demand	$F_6 = \alpha_3 F_2$	$F_1 \leq S$
	$F_7 = F_5 + F_6$	

Table 6: Chemical complex and data equations-Example 2

Unit	MTTF	$C_{prev,j}$	$C_{corr,j}$	α
1	12000	40000	200000	0.92
2	15000	40000	200000	0.85
3	100000	30000	100000	0.75

Table 7: Reliability and operating characteristics-Example 2

$d_1 = 9.813$	$d_2 = 9.028$	$d_3 = 6.187$
$R_{th,1} = .917$	$R_{th,2} = .926$	$R_{th,3} = .968$
Initial Maintenance Sequence: 121213212121321212132111212312121231212132121		
Optimal Maintenance Sequence: 121231212123121213122112312121321212132121		
$N_A = 19$	$N_B = 17$	$N_C = 6$
Expected Profit = 28.84E6		
Expected Maintenance Cost = 2.2E6		

Table 8: Results of the sequential approach-Example 2

$d_1 = 6$	$d_2 = 5.52$	$d_3 = 10$
$R_{th,1} = .893$	$R_{th,2} = .904$	$R_{th,3} = .975$
Initial Maintenance Sequence: 123121231212312132121312312132121		
Optimal Maintenance Sequence: 123121213212312132121312312123121		
$N_A = 14$	$N_B = 13$	$N_C = 7$
Expected Profit = 29.02E6		
Expected Maintenance Cost = 1.8E6		

Table 9: Results of the simultaneous approach-Example 2

State	Sequential	Simultaneous
111	1645	1623
011	577	970
101	577	970
110	1008	593
001	577	970

Table 10: Expected revenue rates-Example 2

approaches are applied to the process shown in Fig. 4, comprising three units with five structurally operable states.

The objective is to find the optimal design and the optimal maintenance policy maximizing an expected profit over a time horizon H. A time horizon of H=20000 has been considered while the cost of raw material A and and the revenue rate of end product C remain as in the previous example. The variables and parameters, along with the equations used to model this system are shown in Table 6. The operating and maintenance characteristics of the process are shown in Table 7.

Table 8 summarises the results obtained by applying the three steps of the sequential procedure, while Table 9 summarises the results of the simultaneous procedure. A summary of the expected revenue rates for the five operable states of the two approaches is presented in Table 10.

Note (by comparing Tables 8 and 9) that the two approaches results in conceptually different designs. The sequential approach directs the route of production via units 1 and 2 while for the simultaneous approach unit 3 is the key unit. This is also evident in the expected revenue rate of the operable system states (Table 10). Note also that in the design determined by the sequential approach, states with units 1 and 2 operable are more profitable than states with unit 3 operable; the opposite occurs in the design determined by the simultaneous approach. This is attributed to the fact that the sequential approach does not consider any reliability and maintenance aspects at the design phase and solely bases its design selection upon process char-

acteristics (conversion factors and structure). On the other hand, the simultaneous approach balances process, reliability and maintenance characteristics while selecting an optimal design. As a result, it directs production via unit 3 which has a smaller conversion factor (α=0.75) but is much more reliable and cheaper to maintain. Over the time horizon, this compensates for production losses in the short term.

Finally, note that the optimal maintenance policy given by the simultaneous approach requires less maintenance actions since it explicitly accounts for long term maintenance costs. This implies that an optimal design is selected in order to facilitate, amongst other things, a less expensive maintenance policy. In the sequential approach, on the other hand, maintenance policy seems to have the role of correcting design misjudgements in order to keep the process going. As a result, higher maintenance costs are incured, resulting in lower (over the time horizon) expected profits.

NOMENCLATURE

M	= the number of components in the system
ER_H	= the expected revenue of the process throughout the time horizon
ER_k	= the expected revenue per time unit from state k of the system
EP_H	= the expected profit of the process throughout the time horizon
d	= the vector of design variables
z	= the vector of control variables
x	= the vector of design variables
θ	= the vector of uncertain parameters of the process
S	= the set of all states of the system
FOR_k	= the feasible operating region of state k
N	= the number of maintenance actions
t_i	= the time instant of $(i-1)^{th}$ maintenance action (continuous variable)
$u_{i,j}$	= integer variable takes the value of 1 if $(i-1)^{th}$ maintenance action is performed on unit j (0-1 variable)
$R_{i,j}(t)$	= the probability of unit j being up after $(i-1)^{th}$ maintenance action at time t
$\bar{R}_{i,j}(t)$	= the probability of unit j being up during $(i-1)^{th}$ maintenance action at time t
$P_{i,k}(t)$	= the probability of the system being in state k after $(i-1)^{th}$ maintenance action at time t
$\bar{P}_{i,k}(t)$	= the probability of the system being in state k during $(i-1)^{th}$ maintenance action at time t
$C_{prev,j}$	= the preventive maintenance cost for unit j
$C_{corr,j}$	= the corrective maintenance cost for unit j
τ_j	= the duration of maintenance action performed on unit j
$\mathcal{R}_{th,j}$	= reliability threshold value of unit j
$MTTF_j$	= the mean time to failure of component j.
α	= the mass balance coefficient

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